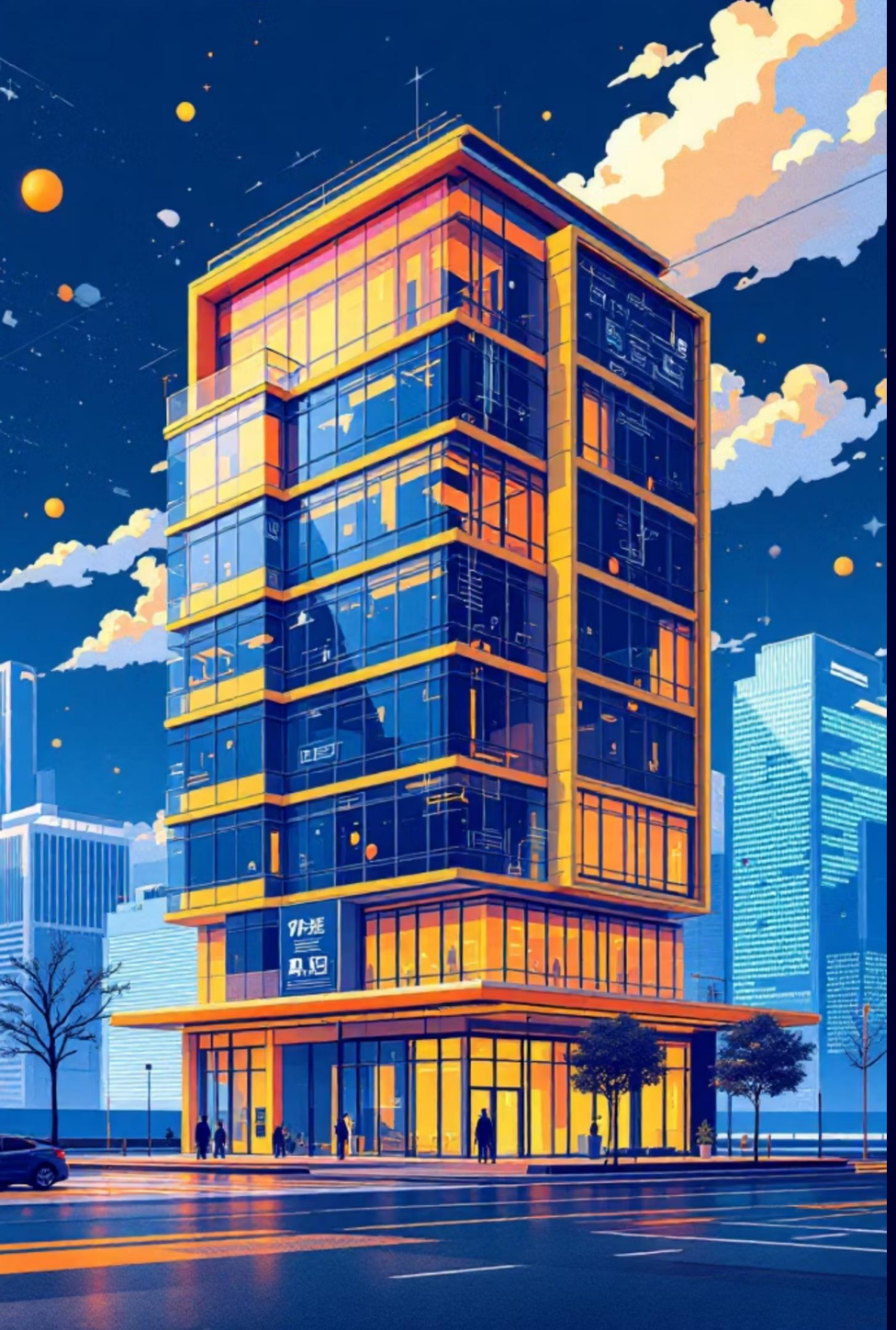


Employee Group Insurance Management System

A Modern, Automated Enterprise Solution for Corporate Insurance

Tech Stack: .NET 8, Angular 20+, SQL Server

- **V. Hemanth Kumar**
  Intern, The Hartford

Transforming Corporate Insurance Operations

The Challenge

Manual insurance processing creates bottlenecks in corporate environments. Premium calculations are error-prone, claim settlements take weeks instead of days, and employees lack visibility into their coverage status.

Our Solution

A scalable, multi-tenant web application that automates the entire insurance lifecycle—from corporate client onboarding to automated claim settlement with real-time transparency.

Comprehensive User Roles & Access Control

System Administrator

Manages insurance plans, master data configurations, and high-level system settings across all corporate clients.

Insurance Agent

Serves as the bridge between system and corporate clients, handling client onboarding and policy management workflows.

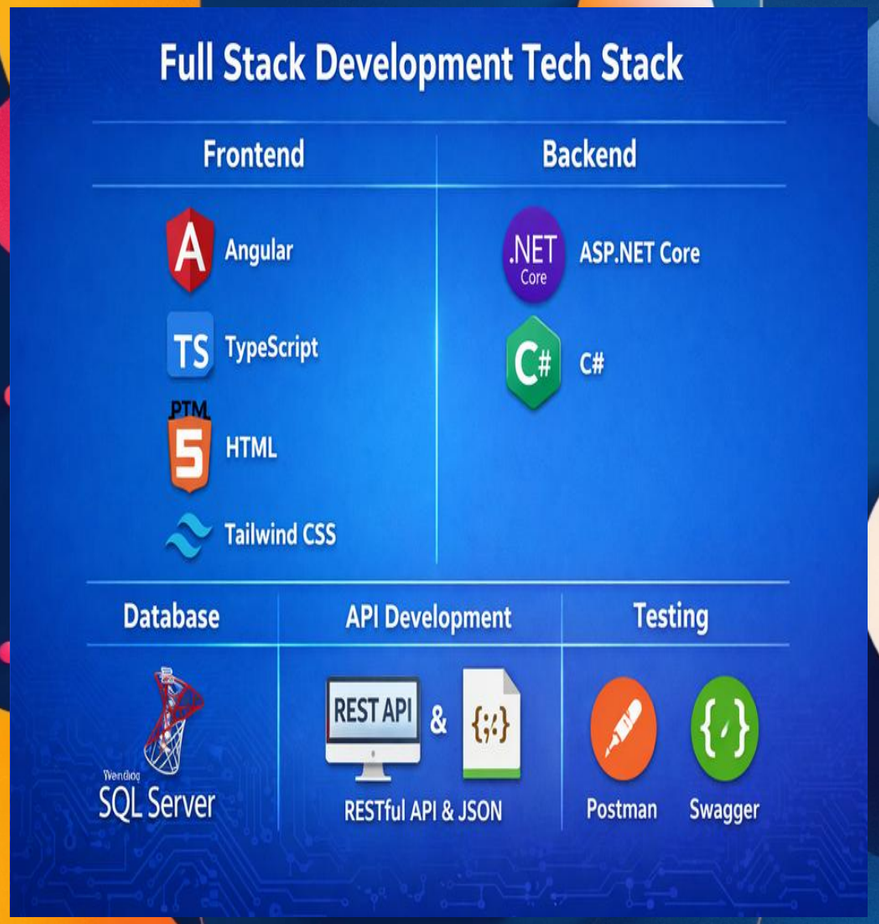
Corporate Client

Manages employee rosters, reviews invoices, and initiates claims for their organization's insurance coverage.

Claims Officer

Expert medical adjudicator who reviews clinical evidence and documentation to settle claims accurately.

Security is implemented using JWT Bearer Tokens with Role-Based Access Control (RBAC) ensuring data protection and compliance.



Enterprise-Grade Backend Technology Stack

.NET Core 8 Web API

Modern framework delivering high-performance RESTful endpoints with built-in support for asynchronous operations and minimal latency.

Entity Framework Core

Code-First approach with 40+ migration scripts ensuring database schema evolves with business requirements while maintaining data integrity.

SQL Server

Enterprise-grade relational database providing ACID compliance, full-text search, and robust backup/recovery capabilities.

Modern Frontend Architecture

Angular Framework

Latest Angular version delivers component-based architecture with modular design patterns, lazy-loaded routes for instant navigation, and Ahead-of-Time (AOT) compilation for production optimization.

State Management

Angular Signals combined with RxJS provides reactive programming model, enabling real-time UI updates as data changes while maintaining predictable state management.

Premium Styling

- Tailwind CSS for utility-first styling approach
- Custom Glassmorphic design elements
- Responsive layouts for all device sizes
- Accessibility-compliant color schemes

Clean Architecture Implementation

Domain Layer

Contains enterprise entities, value objects, enums, constants, and business rules that represent core insurance concepts independent of infrastructure.

Application Layer

Implements business logic, data transfer objects (DTOs), AutoMapper configurations, and validation rules orchestrating domain operations.

Infrastructure Layer

Manages database context, repository implementations, background worker services, and third-party integrations through abstraction.

Web API Layer

Exposes RESTful endpoints consumed by frontend, implementing controllers, middleware, and HTTP request/response handling.

Design patterns include Repository Pattern for data access abstraction, Dependency Injection for loose coupling, and Unit of Work for transaction management.

Database Schema & Entity Relationships

1

Corporate Client

Organizational entity with business classification and billing information

2

Policy Assignment

Links clients to insurance plans with coverage terms and premium rates

3

Members & Dependents

Employee roster with family tree relationships and coverage entitlements

Financial Transactions

Policy links to invoices and payments with partial/full payment support and reconciliation workflows.

Claim Evidence

Claims connect to supporting documents including medical bills, prescriptions, and diagnostic reports.

Audit Trail

Comprehensive AuditLogs table captures every major action with timestamp, user, and change details.

Corporate Client Onboarding Workflow

01

KYC Document Collection

Corporate clients upload GST certificates, PAN cards, incorporation documents, and business licenses through secure portal.

02

Agent Submission

Insurance agents validate document authenticity, complete due diligence, and submit complete application package to administration.

03

Admin Approval Process

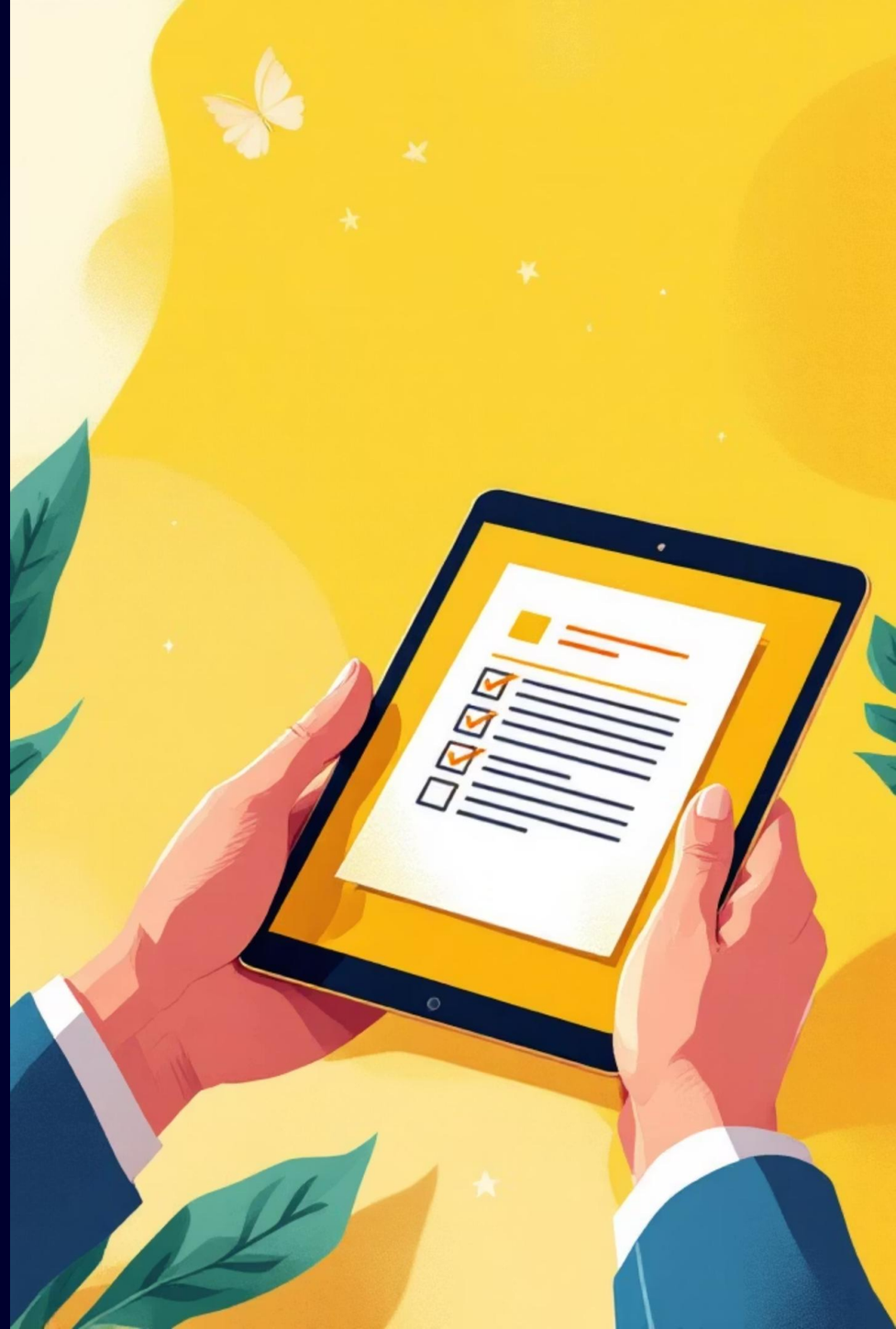
System administrators review documentation, verify business eligibility, and approve or reject applications with detailed feedback.

04

Dashboard Activation

Upon approval, corporate dashboard activates with full functionality, policy configuration options, and employee management capabilities.

Scalability: Supports diverse business categories (Small, Medium, Large, Enterprise) with custom commission rate structures for agent partnerships.



Plan & Policy Management Capabilities

Modular Coverage Plans

Health, Life, and Accident insurance coverages are bundled into customizable plans. Each plan defines coverage limits, exclusions, co-pay percentages, and network provider relationships.

Dynamic Premium Calculation

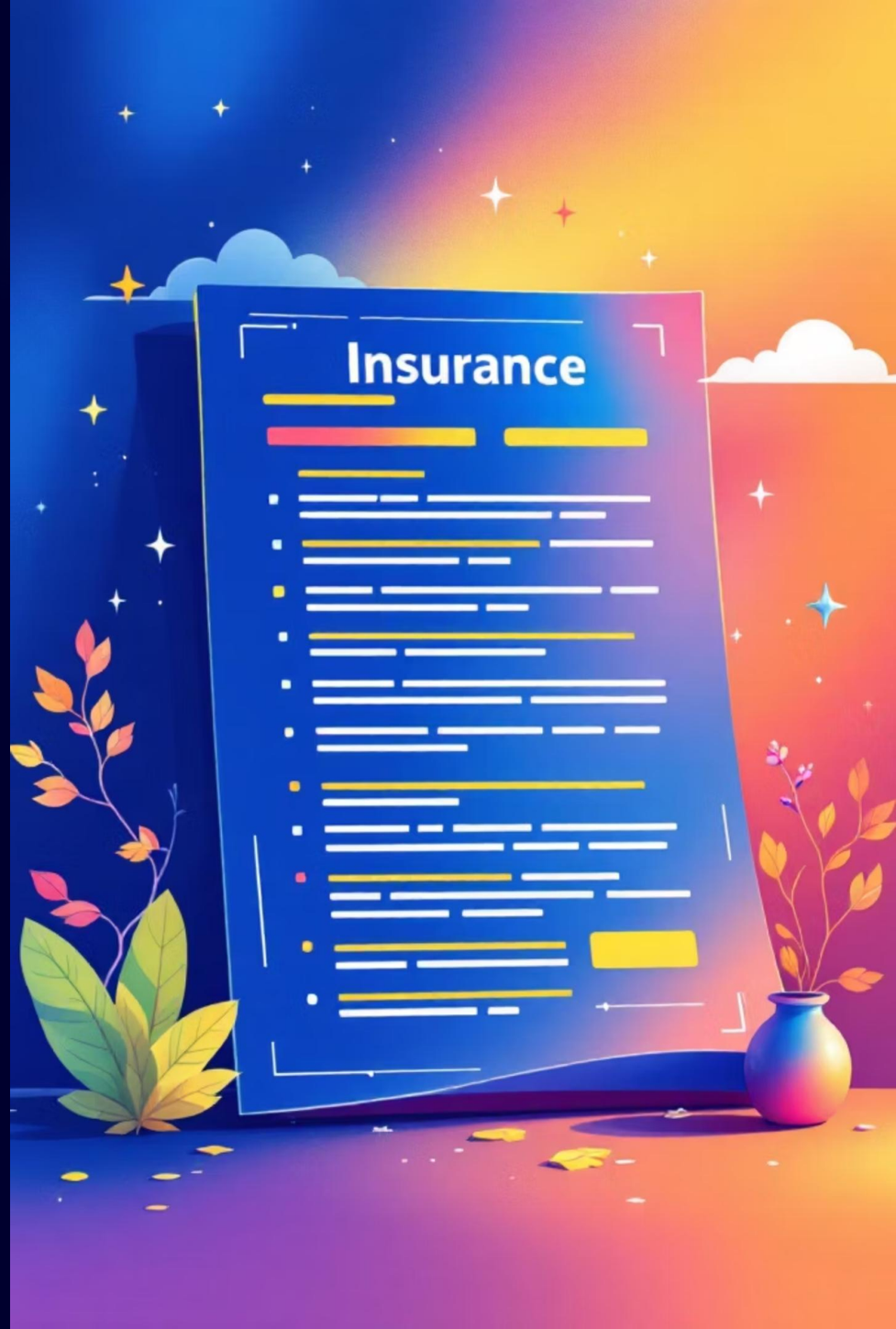
Automated premium calculations based on employee count, age bands, geographic location, and coverage tier selections with real-time cost estimates.

Policy Lifecycle Management

Handles policy start dates, end dates, renewal reminders, and grace period management with automated notifications and escalation workflows.

Mid-Term Endorsements

Supports policy revisions including adding or removing members, changing coverage levels, and updating beneficiary information without policy cancellation.



Member & Dependent Management

Bulk Employee Management

Corporate administrators can onboard hundreds of employees per client through CSV imports or API integrations, with validation rules ensuring data quality.

Family Tree Structure

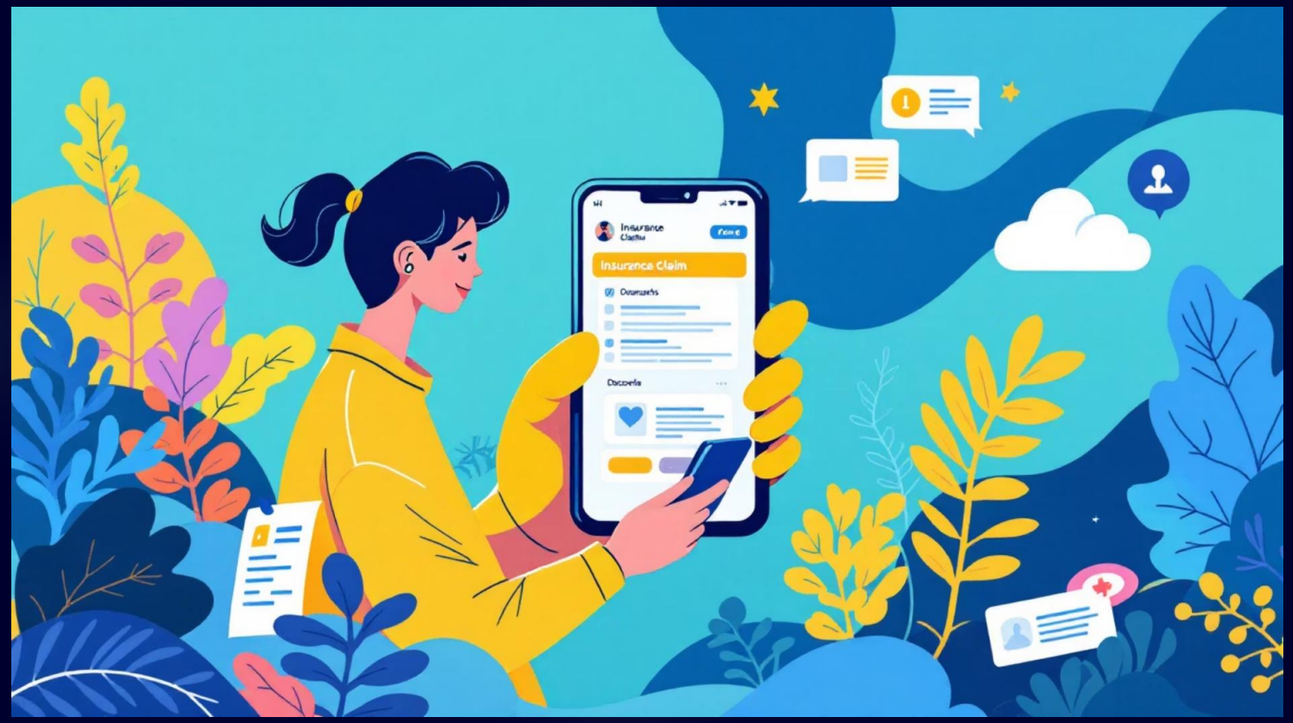
Primary employees establish relationships with dependents including spouse, children, and parents, creating comprehensive coverage maps with inheritance rules.

Real-Time Entitlement Tracking

Each member maintains running balance of "Sum Insured" coverage, deductibles, and co-pay amounts updated with every claim transaction.

System supports hierarchical dependent relationships, coverage eligibility rules based on employment status, and automatic entitlement recalculations when policy changes occur.

Digital Claim Submission



Seamless Member Experience

Empowering members with self-service capabilities that eliminate paperwork and accelerate claim initiation.

Self-Service Portal

Members file claims directly through their personalized dashboard, reducing administrative burden and processing time.

Secure Evidence Upload

PDF and image medical reports uploaded to specialized system with encrypted storage and access controls.

Real-Time Validation

Automatic policy balance verification before submission prevents invalid claims and improves first-time approval rates.

Adjudication Engine

Streamlined claims processing interface giving officers complete visibility and one-click decision capabilities with instant balance updates.

01

Claims Officer Queue

Dedicated dashboard displays all pending claims sorted by priority and submission time.

02

Side-by-Side Comparison

Side-by-side view of requested amount versus remaining policy limit enables informed decisions.

03

One-Click Decisioning

Approve or reject with documented reasons, triggering instant policy balance updates for members.

Background Service Automation

InsuranceAutomationWorker

Continuous 24/7 background processing ensuring financial operations run seamlessly without manual intervention.

Critical Automation Tasks

The worker service handles time-sensitive operations including monthly billing generation 30 days in advance, automatic policy termination at midnight on end dates, and blocking new claims for overdue accounts.

Auto-Billing

Monthly invoices generated 30 days in advance with grace period calculations.

Auto-Expiration

Coverage automatically terminates at midnight on policy end date.

Safety Logic

Blocks new claims for clients with overdue invoices.

Analytics & Dashboards

Real-time insights across operational efficiency, financial performance, and individual member utilization for data-driven decision making.

Efficiency Metrics

Average claim processing time tracked in days, identifying bottlenecks and optimization opportunities across the adjudication workflow.

Financial KPIs

Approved amount versus rejected amount graphs for the insurance provider, enabling financial forecasting and risk assessment at aggregate levels.

Member Tracking

Claim history and policy limit utilization bars for individual employees, providing visibility into usage patterns and remaining coverage balances.

Security & Data Integrity

Enterprise-grade security measures ensuring data protection, graceful shutdown handling, and precision financial calculations.

Graceful Shutdown

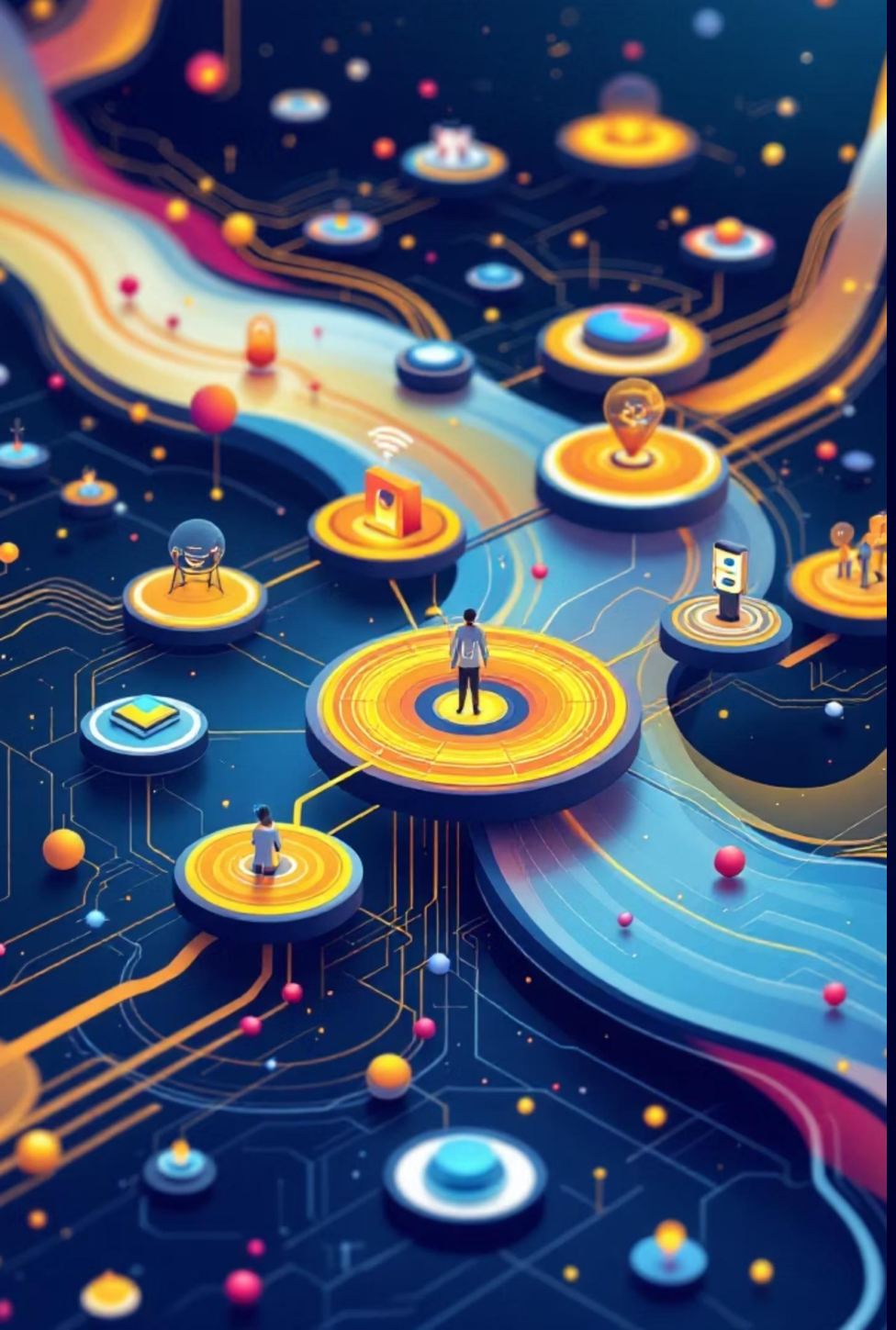
CancellationToken support in background tasks prevents data corruption during service restarts or deployment updates.

Environment Safety

Sanitized .gitignore prevents sensitive configuration and upload files from accidental exposure to version control repositories.

Precision Handling

High-precision decimal(18,2) data type eliminates floating-point errors in financial calculations and reporting.



Future Roadmap

Next-generation capabilities expanding platform functionality with AI-driven fraud detection, mobile accessibility, and integrated payment processing.

1. **AI-Based Fraud Detection**

   Machine learning algorithms analyzing claim patterns to identify suspicious activity and reduce fraudulent submissions.

2. **Mobile App**

   Cross-platform mobile application enabling claims submission and policy management from iOS and Android devices.

3. **Payment Gateway Integration**

   Direct integration with Stripe and Razorpay for seamless premium collection and invoice payments.

Acknowledgements

I would like to express my sincere gratitude for the incredible support, professional growth, and technical training I have received during the development of this project.

The Hartford

Thank you for providing the opportunity and the platform to build meaningful solutions for the insurance sector.

Surya Thammiraju

Managing Director and India Head

My sincere thanks for the leadership, vision, and encouraging environment that allows us to innovate and grow.

GUVI

Thank you for the comprehensive technical training and resources that bridged the gap between concept and real-world application.

Ranjan Bhatnagar

Microsoft Certified Trainer

I am deeply grateful for your constant mentorship, technical expertise, and dedicated support as our trainer throughout this program.

Koti Rama Murthy Davuluri

Senior Staff Software Engineer at The Hartford

I am deeply grateful for your technical guidance, support, and for always being a mentor during the development phase.

Thank you!